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Question 1

1. `private f()`
2. 不能, 因为子类不能缩小覆盖方法的权限, 即不能 `private` 覆盖 `public`

Question 2

1.

```
sup.field = 0, sup.getField() = 1
sub.field = 1, sub.getField() = 1,
sub.getSuperField() = 0
```

2. 不能, 因为没有发生覆盖, 而只是隐藏

Question 3

1.

```
Base staticGet()
Derived dynamicGet()
```

2. 不能, 因为静态属性和静态方法都不会发生覆盖, 只是发生隐藏

Question 4

1. 不能通过编译, 因为方法 `study` 没有发生覆盖, 父类中没有, 无法通过父类进行调用

改动方法: 在父类中加入 `study()` 方法的被覆盖代码, 则子类会覆盖父类的方法, 父类也就可以进行调用了

```
public void study() {
    System.out.println("父类studying");
}
```

2.

```
parent.name=李四
child.name=张三
子类eating
父类sleeping
子类eating
子类sleeping
子类studying
```

3. 对于第1个输出 `parent.name=李四`：非静态变量发生隐藏，向上转型之后输出父类的结果

对于第2个输出 `child.name=张三`：子类输出自己的变量结果

对于第3个输出 `子类eating`：方法发生覆盖，向上转型之后仍然输出实例化对象的方法调用结果

对于第4个输出 `父类sleeping`：父类中该方法为 `static`，无法发生覆盖，只能是隐藏，在向上转型之后输出父类方法的结果

对于第5个输出 `子类eating`：子类调用自身的方法

对于第6个输出 `子类sleeping`：子类调用自身的方法

对于第7个输出 `子类studying`：方法发生覆盖，向上转型之后仍然输出实例化对象的方法调用结果

Question 5

见代码文件 `Question05`

Question 6



The screenshot shows the IntelliJ IDEA IDE with a project named "Question06". The source code is in a file named "Test.java". The code defines two classes: `Outer` and `Inner`. `Outer` has a public static integer `num` set to 10. `Inner` is a non-static inner class that has its own public integer `num` set to 20. The `show()` method in `Inner` prints the value of `num` (which is 20), then prints `this.num` (which is 20), and finally prints `Outer.this.num` (which is 10). The `Test` class has a `main` method that creates an instance of `Outer.Inner` and calls its `show()` method. The output window shows the execution results: 30, 20, and 10. The status bar at the bottom indicates the file is "Test.java" and the encoding is "UTF-8".

```
1 class Outer {
2     public int num = 10;
3
4     class Inner {
5         public int num = 20;
6
7         public void show() {
8             int num = 30;
9             System.out.println(num);
10            System.out.println(this.num);
11            System.out.println(Outer.this.num);
12        }
13    }
14 }
15
16 public class Test {
17     public static void main(String[] args) {
18         Outer.Inner oi = new Outer().new Inner();
19         oi.show();
20     }
21 }
```

运行 Test

"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:J:\IntelliJ IDEA 2024.1\lib\idea_r...

30
20
10

进程已结束，退出代码为 0

Question 7



Question 8

见代码文件 *Question08*

思路：

首先初始化 `ShapeSequence` 实例化对象，初始化大小为 **6**，之后加入多个实例化的形状变量，输出检测

实例化一个 `Iterator` 迭代器对象，输出当前形状

不断移动，直至移动到最尾部，即无法继续移动为止，判断是否已经到尾

再创建第二个 `Iterator` 迭代器对象，进行移动，并在移动的开头和结尾进行比较这两个对象指向的地址是否 `equals`

代码：

```

1 public class Test {
2     public static void main(String[] args) {
3
4         ShapeSequence a = new ShapeSequence(1000);
5         a.add(new Rectangle(x = 10, y = 10));
6         a.add(new Rhombus(x = 10, y = 10));
7         a.add(new Ellipse(x = 10, y = 10));
8         a.add(new Rectangle(x = 10, y = 10));
9         a.add(new Rhombus(x = 10, y = 10));
10        a.add(new Ellipse(x = 10, y = 10));
11        System.out.println(a);
12
13        ShapeSequence.Iterator it = a.iterator();
14        System.out.println("current? " + it.current());
15
16        it.moveNext();
17        it.moveNext();
18        it.moveNext();
19        it.moveNext();
20        it.moveNext();
21        System.out.println("isEnd? " + it.isEnd());
22
23        ShapeSequence.Iterator it2 = a.iterator();

```

```

26 System.out.println("equals? " + it.equals(it2));
27
28 it2.moveNext();
29 it2.moveNext();
30 it2.moveNext();
31 it2.moveNext();
32 it2.moveNext();
33 System.out.println("equals? " + it.equals(it2));
34
35 System.out.println("isEnd? " + it2.isEnd());
36
37 }
38
39 }

```

输出：

The screenshot shows the IntelliJ IDEA IDE with a Java file named 'Test.java' open. The code defines a 'Rectangle' class with a 'current' attribute and a 'move' method. The 'move' method updates the 'current' attribute based on the 'x' and 'y' parameters. The IDE interface includes a toolbar at the top, a project explorer on the left, and a status bar at the bottom showing 'QuestionDB > src > Test > main' and '35.31s CRLF UTF-8 4 个字节'.

```

1  package com.questiondb;
2
3  import java.awt.geom.*;
4
5  public class Rectangle {
6      private Point current;
7
8      public Rectangle(Point p) {
9          current = p;
10     }
11
12     public void move(int x, int y) {
13         current = new Point(x, y);
14     }
15 }

```